

[ARCHIVED CATALOGUE]

## Electrical Engineering (VLSI Design) (MS)

---

The Master of Science in Electrical Engineering (VLSI design) is earned by successfully completing the normal requirements for the Master of Science in electrical engineering, with the following additional required courses: [EE 536a](#); [EE 577a](#); [EE 577b](#) or [EE 536b](#); and [EE 552](#). If a student chooses to take the first of the OR courses, the other course may be counted in the VLSI/CAD, Circuits and Devices and Computer Systems areas as indicated, below.

For Admission Requirements, refer to [Viterbi Graduate Degrees and Requirements](#).

No more than three courses (maximum 12 units) may be counted at the 400 level — at least 19 adviser-approved units must be taken at the 500 or 600 level.

The students must also take two courses from one of the following areas and one course from a second area:

### VLSI/CAD Area

---

- [CSCI 455x Introduction to Programming Systems Design](#) Units: 4
- [EE 554 Cyber-Physical Systems: A Computing Perspective](#) Units: 4
- [EE 560L Digital System Design](#) Units: 4
- [EE 576 Integrated Memory Devices and Technologies](#) Units: 4
- [EE 578 Computational Electromagnetics for Engineers](#) Units: 4
- [EE 580 System Verification](#) Units: 4
- [EE 581 Mathematical Foundations for System Design: Modeling, Analysis, and Synthesis](#) Units: 4
- [EE 631 Mixed-Signal Integrated Circuits](#) Units: 4
- [EE 658 Diagnosis and Design of Reliable Digital Systems](#) Units: 4
- [EE 680 Computer-Aided Design of Digital Systems](#) Units: 4

### Circuits and Devices Area

---

- [EE 448L Communication Electronics](#) Units: 4
- [EE 504L Solid-State Processing and Integrated Circuits Laboratory](#) Units: 4
- [EE 505 Analog, Mixed-Signal, and RF Integrated-Circuit Tape-Out](#) Units: 4
- [EE 536a Analog Integrated Circuits](#) Units: 4
- [EE 536b Analog Integrated Circuits](#) Units: 4
- [EE 537 Modern Solid-State Devices](#) Units: 4
- [EE 582 CMOS:Nano Neuromorphic Circuits](#) Units: 4
- [EE 576 Integrated Memory Devices and Technologies](#) Units: 4
- [EE 631 Mixed-Signal Integrated Circuits](#) Units: 4
- [EE 632a Integrated Communication Systems](#) Units: 4
- [EE 632b Integrated Communication Systems](#) Units: 4

### Computer Systems Area

---

- [CSCI 455x Introduction to Programming Systems Design](#) Units: 4
- [CSCI 570 Analysis of Algorithms](#) Units: 4

- [EE 541 A Computational Introduction to Deep Learning](#) Units: 2
- [EE 557 Computer Systems Architecture](#) Units: 4
- [EE 560L Digital System Design](#) Units: 4
- [EE 568 Parallel Programming](#) Units: 4
- [EE 577b VLSI System Design](#) Units: 4
- [EE 580 System Verification](#) Units: 4
- [EE 595 Software Design and Optimization](#) Units: 4
- [EE 659 Advanced Topics in Interconnection Network Design and Analysis](#) Units: 4
- [EE 677 Accelerated Computing using Field Programmable Gate Arrays](#) Units: 2

## Note:

---

With explicit approval of a faculty adviser, [EE 599 Special Topics](#) and/or 3 units of [EE 590 Directed Research](#) may be used to meet requirements for any of the approved areas.

## The remaining courses must be technical electives approved by the adviser, and can include the following:

---

- [EE 457 Computer Systems Organization](#) Units: 4
  - [EE 477L MOS VLSI Circuit Design](#) Units: 4
  - [EE 501 Solid State](#) Units: 4
  - [EE 502 Advanced Solid State](#) Units: 3
  - [EE 504L Solid-State Processing and Integrated Circuits Laboratory](#) Units: 4
  - [EE 506 Semiconductor Physics](#) Units: 4
  - [EE 540 Introduction to Quantum Electronics](#) Units: 4
  - [EE 554 Cyber-Physical Systems: A Computing Perspective](#) Units: 4
  - [EE 560L Digital System Design](#) Units: 4
  - [EE 590 Directed Research](#) Units: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
  - [EE 601 Advanced Semiconductor Device Physics](#) Units: 4
  - [EE 677 Accelerated Computing using Field Programmable Gate Arrays](#) Units: 2
-